

MERGE 4 DESIGN LLC × L.E.E.P., INC.

FORWARD HOME PROGRAM

Project Development Document

Proposed Building Package for the City of Wilmington Vacant Lot Initiative

\$220K–\$250K T1 target sale price	4–10+ units Per lot cluster	31 weeks Design to CO	4 tiers available Code+ → Off-Grid Harvest
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Merge 4 Design LLC A Wilmington Delaware Mission Driven Partnership merge4design.com Predevelopment · Architecture · Cost Intelligence	L.E.E.P., Inc. Gwenevere Motley, Executive Director Local Economic Empowerment Program Trained Forward Home installation workforce
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May 2026 · Companion files: M4D_ForwardHome_BuildingSpec_May2026.docx · M4D_ForwardHome_Proforma_May2026.xlsx

1. Executive Summary

Merge 4 Design LLC and L.E.E.P., Inc. propose to deliver complete Forward Home building packages on City of Wilmington vacant lots across four escalating performance tiers — from code-compliant forward housing at \$220K–\$250K through net-zero, net-positive, and fully off-grid homes with agrivoltaic food-producing rooftop gardens. All four tiers share the same Forward Panel wall system and delivery infrastructure. L.E.E.P., Inc. provides the trained installation workforce throughout.

The central quality claim in this document is not marketing — it is verifiable. The 4Ward-SIB Forward Panel assembly costs less per square foot than the equivalent stick-frame material assembly it replaces, before a single labor saving, schedule compression, or trade elimination is counted. The national average HERS score for new US production homes in 2024 was 55. The Forward Home Tier 1 matches that standard automatically via SIP construction at a \$220K–\$250K forward housing price that no production builder delivers anywhere in the United States today.

These homes are not the affordable version of something. They are the forward version — built to a standard that immediately sets a new comparable in every neighborhood where they land.

KEY FIGURES (10-unit Type D cluster): Build cost ~\$140K/unit · T1 sale price \$220K–\$250K · T4 off-grid sale price \$267K–\$317K · Developer margin ~40% (Type D) across all configurations · Build schedule 31 weeks · Land: \$1 (City program) · Site work + utilities: City-funded · L.E.E.P. workforce: all four tiers

2. Why This Quality Level at This Price

2A. The national context

The average new production home built in the United States today is clad in vinyl siding (now the most-used primary exterior wall material in the country, having surpassed stucco per US Census/NAHB data), sheathed in OSB with a separate house wrap, and delivers a national average HERS score of 55 (RESNET 2024 — 33% of all new US single-family homes rated). To be considered high-performance,

a home must score HERS 70 or below. KB Home, the best-performing national production builder by HERS, averaged HERS 45 in 2024.

Builder-grade windows, the standard in production affordable housing, are specified to pass minimum energy code, not to perform. Standard stick-frame affordable housing delivers 5.0–7.0 ACH50 blower door results and no CI layer.

2B. What Forward Home delivers at the same price

The Forward Home Tier 1, at \$220K–\$250K, delivers:

- **Non-combustible structural wall panel** (cellulosic fiberboard — not OSB, not vinyl, not combustible) with integrated WRB and R-5 GPS foam CI in a single product.
- **Whole-wall R-21 to R-23 effective** — compared to R-18 effective for a standard 2×6 R-23 cavity-only stick frame wall, due to thermal bridging at studs. The panel eliminates the thermal bridge.
- **Crystal Magnus 4500 tilt-turn windows** at $U \leq 0.20$ — a European-profile uPVC window with multi-point locks, dual compression seals, PG70 structural rating, and STC attenuation to 41 dB. Production affordable housing uses builder-grade single-hung vinyl at $U \leq 0.32$.
- **Allura fiber cement cladding** with 50-year limited warranty. Not vinyl. Not painted wood. Non-combustible, moisture-resistant, impact-resistant.
- **Energy Star automatic** through SIP construction — no blower door test required for Energy Star qualification. Stick-frame affordable housing rarely achieves Energy Star without additional cost measures.
- **30-year warranted panel system** with TER code-evaluation documentation for every AHJ in the United States.

THE DEFENSIBLE CLAIM: No production builder in the Mid-Atlantic, and we believe in the United States, is delivering a non-combustible, R-21+ effective wall, SIP-panelized, Energy Star automatic, tilt-turn window assembly at \$220K–\$250K in any market today. The first project proving it is in active construction in Wilmington (AE24-002, 1316–1324 N. Claymont St., foundation permit approved May 6, 2026, panels arriving June 22, 2026).

2C. Neighborhood value lift

These homes are priced at the low tier of the current market. That gap between price and quality has a consequence: when a cluster of Forward Homes closes at \$220K–\$250K in a Wilmington neighborhood where the existing comp base is distressed, appraisers will initially struggle to find comparable sales at that quality level. Within 12–24 months of first occupancy, closed sales data from Forward Homes will begin to establish new comparables. Assessed values in the surrounding blocks will reflect the new standard.

This is not speculative; it is the documented mechanism behind every LIHTC and NMTC investment thesis in US affordable housing policy. New high-quality construction in underinvested neighborhoods raises comparable sales. The Forward Home system accelerates this process because the quality gap between a Forward Home and surrounding existing stock is larger than in any comparable program.

Once the first cluster is complete and comparables are established, the upgrade tiers are not aspirational, they are warranted by the market the first homes created. A Tier 2 or Tier 3 home in the same neighborhood, 12 months later, is priced against a comp base that Forward Homes themselves established. That is the system play.

3. The Forward Panel Cost Argument

3A. The wrong comparison

The comparison that makes panelized systems appear expensive is panel material cost versus OSB sheathing alone — approximately \$3–5/SF versus \$0.30–0.60/SF. That comparison is wrong because OSB is one layer of a four-layer code-compliant assembly. The Forward Panel replaces all four layers in one step.

3B. Equivalent assembly comparison — layer by layer

Wall assembly component	4Ward-SIB Forward Panel	Standard SIP (OSB face)	Stick frame (all layers)	Panel saving vs. stick
Structural sheathing	✓ Included	OSB ~\$2.00– 3.00/SF	\$2.00–3.00/SF	
Weather-resistive barrier (WRB)	✓ Included	Separate ~\$0.80–1.20/SF	\$0.80–1.20/SF	
Air infiltration barrier (AIB)	✓ Included	Separate ~\$0.30–0.50/SF	\$0.30–0.50/SF	
R-5 continuous insulation (CI)	✓ Included	Separate ~\$1.50–2.50/SF	\$1.50–2.50/SF	
Cavity insulation sub labor	✓ Eliminated	Separate ~\$1.00–1.50/SF	\$1.50–2.50/SF	
Panel engineering	~\$0.90–1.20/SF	~\$1.40–1.80/SF	—	
MATERIAL TOTAL (per SF of wall)	~\$3.00–5.00/SF	~\$7.00– 10.50/SF	~\$7.10– 11.20/SF	\$2.10–6.20/SF saved

Sources: Panel material estimates from M4Di ALDDB and Barricade INDEVCO market data (May 2026). Stick frame assembly costs per RS Means 2025 Residential Cost Data. House wrap \$0.80–1.20/SF per Owens Corning installer data. CI \$1.50–2.50/SF per Polyiso Manufacturers Association 2025. All figures per SF of exterior wall surface area.

THE BOTTOM LINE: The 4Ward-SIB Forward Panel system costs \$3.00–5.00/SF for the complete four-layer assembly. The equivalent stick-frame assembly costs \$7.10–11.20/SF. The panel is less expensive before any labor saving, schedule compression, or HVAC downsizing is counted.

3C. Labor and schedule savings

Panelized construction labor advantages are documented by independent research:

- **55%–66% framing labor reduction** versus conventional stick frame. Source: RS Means Residential Cost Data study commissioned by the Structural Insulated Panel Association (SIPA).
- **4-person L.E.E.P. crew installs ~100 linear feet of wall panel in under 3 hours** — replacing four separate subcontractor mobilizations with one.

3D. Schedule savings — carrying cost in real dollars

Build method	Schedule	Weeks saved vs. stick	Loan carry saved (10-unit @10.25%)	Per unit
4Ward-SIB Forward Panel	31 weeks	8 weeks	~\$52,000	~\$5,200
Standard SIP (OSB face)	35 weeks	4 weeks	~\$26,000	~\$2,600
Stick frame (conventional)	39 weeks	—	Baseline	—

Source: Build schedule estimates from Offsite Builder Magazine 2025. Carrying cost: \$260,000 loan × 10 units × 10.25% annual ÷ 52 weeks × weeks saved.

3E. HVAC downsizing savings

- **1–2 ton HVAC unit downsizing** per unit in a panelized assembly versus equivalent stick-frame home. Source: ACCA Manual J; SIPA DOE Building America study.
- **\$3,000–\$8,000 equipment savings** per unit at current Delaware heat pump installed cost rates.
- **\$1,800–\$3,200 annual energy savings** per unit versus comparable stick-frame home. Source: DOE Building Science Corporation, SIP energy performance studies.

3F. Fixed vs. volatile pricing

Stick-frame construction is priced against the lumber futures index, which demonstrated 40%–300% price swings between 2019 and 2025. The 4Ward-SIB Forward Panel face material is cellulosic fiberboard — not wood, not subject to the lumber index. Barricade states explicitly that SIB pricing is stable relative to OSB market fluctuations. For a fixed-price commitment to the City of Wilmington, this pricing stability is a structural risk reduction that does not appear in a per-SF comparison but is real and consequential.

4. Four-Tier Forward Home System

	Tier 1 Code+	Tier 2 Zero	Tier 3 Positive	Tier 4 Harvest
HERS index	55–70	0–30	Neg to +10	Net positive
Blower door (ACH50)	< 2.0 auto	< 1.0	< 0.6	< 0.6
Whole-wall R	R-21–23	R-22–24	R-24–26	R-24–26
HVAC	Std mini-split ≥ SEER2 18	Quilt 3-zone SEER2 25.3	Mitsubishi FX SEER2 29.9	Quilt battery-opt. SEER2 25.3
Smart panel	200A stub	SPAN MAIN 40 UL 3141	SPAN MAIN 40	SPAN MAIN 40 Island mode
PV	Stub only	5.5 kW	8.5 kW	8.5–12 kW agrivoltaic

Battery	—	Enphase IQ5P or Powerwall 3	Enphase IQ5P or Powerwall 3	EcoFlow DPU + FranklinWH 90kWh
Rooftop	Std membrane	LiveRoof® Standard	LiveRoof® MAXX + edible rows	Crop Circle Farms® agrivoltaic trellis
Grid dependency	Grid	Grid	Grid (PV export)	Off-grid capable
Annual utility cost	\$900–\$1,400	\$200–\$500	~\$0 to credit	\$0 (off-grid)
Food production	—	—	200–400 lbs/yr	200–600 lbs/yr
Certification	Energy Star auto	DOE ZERH	PHIUS+	Off-grid verified
IRA \$45L (developer)	—	\$5,000/unit	\$5,000/unit	\$5,000/unit
Sale price (net \$45L)	\$220K–\$250K	\$227K–\$263K	\$243K–\$287K	\$267K–\$317K
Premium over T1	—	~\$12K–\$18K	~\$28K–\$42K	~\$52K–\$72K

Full tier specification in companion document: *M4D_ForwardHome_BuildingSpec_May2026.docx*

4A. Why four tiers, not one

The same Forward Panel wall system, the same L.E.E.P. installation workforce, and the same M4Di predevelopment cost infrastructure deliver all four tiers. Tier selection is a predevelopment decision made before schematic design. The City of Wilmington lot program operates at Tier 1. Mid-market and market-rate applications of the Forward Building System operate at Tiers 2–3. Luxury and off-grid applications operate at Tier 4. The system scales without changing its core platform.

The upgrade tiers are not a future aspiration — they are documented, product-specified, and priced in the companion spec and proforma. A developer acquiring lots through the City program can commit to Tier 1 at LOI and upgrade to Tier 2 or 3 as the neighborhood comp base rises following first occupancy. The infrastructure to execute the upgrade is already designed in.

4B. Tier 4 — Forward Harvest Off-Grid

Tier 4 is the product that does not exist anywhere in the United States today at any price point delivered by a production system. An off-grid capable, food-producing, agrivoltaic, closed water-energy-food loop home built to Passive House envelope standards and delivered by a trained local workforce — in Wilmington, Delaware, starting at \$267K net to buyer after IRA credits.

- **Quilt 3-zone heat pump** optimized for battery draw — occupancy sensing reduces island-mode energy consumption automatically. SEER2 25.3/HSPF2 12. OTA firmware keeps the system current with no service visit.
- **SPAN MAIN 40 island-mode configuration** — 32 individually controllable circuits; load-shedding priority schedule maintains HVAC, refrigeration, and communications during off-grid operation.
- **EcoFlow DELTA Pro Ultra + FranklinWH aPower 2** — 19.2–27 kWh base, expandable to 90 kWh. True off-grid island mode. No utility interconnect required.
- **Crop Circle Farms® modular rooftop system** — patented, 40+ country trademark. Bifacial TOPCon agrivoltaic PV trellis elevated 5–6 ft above growing beds. Shade-tolerant crops beneath panels; plant transpiration cools underside of panels, improving PV efficiency. 200–600 lbs of produce per unit per year.
- **500-gallon cistern with gravity-fed Rain Bird XFS-CV drip irrigation** — closes the water loop from roof to crop to compost to soil.

- **NatureMill indoor composter** — processes food scraps to soil amendment in 2 weeks at 5W average draw. Feeds the rooftop growing medium. Closes the food waste loop within each unit.

The Tier 4 story is the PHOENIX thesis made physical: a home that continues to produce food and energy regardless of what happens to the grid, the supply chain, or the weather outside. That is a different product category from affordable housing. It is climate-adaptive community infrastructure.

5. Capital Structure and Investor Trigger

Merge 4 Design and L.E.E.P. are not requesting City subsidy beyond the land and infrastructure already committed in the vacant lot program. The capital structure is fully private with one trigger:

- **Site control documentation** — a letter of intent from the City identifying the lot cluster — triggers commitment of private equity from a confirmed investor. The \$500K equity commitment covers three sequential needs: (1) the 20% LTC gap not funded by the construction loan (~\$280K on a Type D cluster), (2) pre-construction soft costs including A&E, CBai-assisted MEP PE fees, permits, and M4Di OI Tool (~\$117K), and (3) a working capital reserve (~\$103K). Soft costs are disaggregated and independently verifiable per M4Di CBai MEP Workflow SOP v1.
- **Construction financing** — site control + investor equity + M4Di ROM verification supports construction loan commitment. Target LTC 80%. Delaware market: 10.0%–11.0%.
- **Panel order** — placed concurrent with permit review, compressing schedule without adding risk.
- **CO and closing** — for-sale units close at CO. M4Di produces final cost report to OED as part of cost-verification program already adopted.

CAPITAL INDEPENDENCE: The Forward Home System and the Merge4Ward manufacturing partnership proceed regardless of this proposal's outcome. Four parallel capital paths are active: (1) grant funding, (2) federal stack (EDA/EPA/HUD/USDA), (3) M4Di operating revenue, (4) private equity. The City is invited to lead a partnership that will exist either way — not asked to underwrite it.

6. Delivery Schedule — Type D 10-Unit Cluster

Phase	Timing	Key deliverables	Milestone
Phase 1 — Predevelopment	Wks 1–6	LOI executed · Investor equity activated · M4Di G0 ROM verification · Site + zoning analysis (Ch. 48) · Tier selection	Equity committed
Phase 2 — Design + Permitting	Wks 7–18	Schematic design · Permit drawings (AOR stamp) · AHJ submission · M4Di ALDi cost control · DOE/PHIUS enrollment (T2–4)	Permit approved
Phase 3 — Fabrication + Site Prep	Wks 12–22	4Ward-SIB panel order placed · Foundation construction · MEP coordination	Panels on site
Phase 4 — Erection + Enclosure	Wks 22–26	L.E.E.P. crew panel install · Dry-in · MEP rough · Windows/doors	Dry-in complete
Phase 5 — Finishes + CO	Wks 26–31	Finishes · MEP trim-out · Inspections · CO · Closings or lease-up	CO issued

7. Responses to Likely Objections

"Panelized construction costs more than stick frame."

Not when you compare equivalent assemblies. The 4Ward-SIB Forward Panel costs \$3.00–5.00/SF for the complete four-layer assembly. The equivalent stick-frame assembly costs \$7.10–11.20/SF. The panel is less expensive on materials before labor savings, schedule compression, or HVAC downsizing are counted. See Section 3 of this document. We will sit down with any builder, cost estimator, or construction consultant and walk through the comparison line by line.

"We don't have crews trained in SIP installation."

That is the point of the L.E.E.P. partnership. L.E.E.P., Inc. provides the trained installation workforce specifically because no existing local workforce is pre-trained in Forward Panel installation. A four-person L.E.E.P. crew can install 100 linear feet of 4Ward-SIB wall panel in under 3 hours. Learning curve is measured in days, not weeks, because the installation is simpler than conventional framing — panel-and-fasten versus measure-cut-nail-wrap-tape-insulate.

"Affordable housing can't absorb a quality premium."

There is no quality premium at Tier 1. The Forward Panel costs less than stick frame per equivalent SF assembled. Where incremental cost does exist at Tiers 2–4, it is offset by: (a) HVAC downsizing credit (\$3K–\$8K/unit), (b) schedule savings (\$5K+/unit at current loan rates), (c) IRA §45L credit (\$5,000/unit at Tiers 2–4), and (d) lower utility costs for residents (\$1,800–\$8,000/year depending on tier). The proforma shows a developer margin of approximately 40% on the Type D 10-unit cluster at the \$235K sale price. This is not a thin deal.

"How do we know these numbers are real?"

M4Di's OI Tool is the City's own adopted cost-verification standard for the A&E Feasibility Grant program (OED letter, May 2026). The same tool OED uses to verify other developers' proposals is the tool M4D uses to verify this one. Soft costs are disaggregated into four independently sourced components — A&E fee, CBai-assisted MEP PE fee (per M4Di CBai MEP Workflow SOP v1), permits, and M4Di OI Tool — not rolled into an opaque lump sum. The verified ROM, including the disaggregated soft cost methodology, is available to the Mayor's office, the Department of Real Estate and Housing, and any lender or investor reviewing this proposal.

"These upgrade tiers are too expensive for Wilmington."

Tier 1 is designed for the City's \$220K–\$250K target. Tiers 2–4 are not Wilmington-program homes — they are the same system applied to mid-market, market-rate, and luxury contexts in any community. Forward Home is a building system, not a housing program. The City program uses Tier 1. The fact that the same platform scales to Tier 4 is a demonstration of system quality, not a proposal to build \$300K homes on City lots.

8. What Merge 4 Design and L.E.E.P. Are Asking For

- **Letter of intent** from the City identifying the ~10-lot cluster for site control.
- **30-day exclusivity** to complete site control documentation and activate investor equity commitment.
- **Expedited AHJ review consideration** for the Forward Panel system (TER 1507-09 and ESR-4635 on file — no novel material review required).

- **Access to the Department of Real Estate and Housing** designated vacant lots to begin formal proposal review.

The City does not write a check. The City does not assume development risk. The City provides land at the committed price (\$1/lot) and site infrastructure already budgeted. Private capital builds the homes. M4Di verifies the numbers. L.E.E.P. provides the workforce. The homes sell or rent within the City's stated target range, without additional subsidy — and they immediately set a new quality comparable for the neighborhoods where they land.

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